SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA 02 **COURSE NAME: Fundamentals of Machine Learning**

COURSE OUTCOME COVERED

CO1: Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory. *(BL3)*

Assessment Tool Weightage:20%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Concept Mapping

Code of Conduct:

I Kruthika Sre S | 192525360 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kruthika

Annexure A

Concept Mapping

Q1 Construct a concept map integrating machine learning foundations with real-world applications such as spam detection or recommendation systems. (5Mark- 7 Min)
Q2 Develop a concept map illustrating how random variables and probability distributions are utilized in predicting uncertain outcomes. (5Mark- 7 Min)
Q3 Formulate a concept map demonstrating the application of Bayes decision theory to medical diagnosis and decision-making problems. (5Mark- 7 Min)
Q4 Design a concept map applying the stages of a learning system to address a student performance prediction problem. (5Mark- 7 Min)
Q5 Synthesize a concept map representing how information theory concepts are employed to quantify uncertainty in machine learning datasets. (5Mark- 7 Min)
Q6 Construct a concept map linking supervised, unsupervised, and reinforcement learning with their practical applications in intelligent systems. (5Mark- 7 Min)
Q7 Develop a concept map illustrating the relationship between random variables, conditional probability, and Bayes theorem in spam classification. (5Mark- 7 Min)
Q8 Formulate a concept map demonstrating how probability distributions support decision-making in weather forecasting systems. (5Mark- 7 Min)
Q9 Design a concept map applying the workflow of a machine learning system to fraud detection in banking transactions. (5Mark- 7 Min)
Q10 Synthesize a concept map representing the role of entropy, information gain, and uncertainty reduction in classification problems. (5Mark- 7 Min)

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Machine Learning

↓ Data → Features → Algorithm → Training → Prediction

↓ Types of Learning

↳ Supervised → Classification / Regression

↳ Unsupervised → Clustering

↳ Reinforcement → Reward-based Learning

↓ Applications

→ Spam detection →

→ Recommendation System →

39/50

2)

Uncertain outcome

↓

Random Variable

↳ Discrete →

↳ Continuous →

↓

Probability Distribution

↳ Assigns probability to each outcome

↓

Probability models

↳ Binomial → success/Failure

↳ Normal → continuous values

↓

Prediction

→ calculate likelihood of outcomes

→ choose most probable outcome

3)

Medical Diagonosis

↓

Patient evidence

Symptoms + Test results

↓

Bayes Decision Theory

↓

Prior Probability + Likelihood of evidence

↓

Bayes Theorem

$$P(D|E) = \frac{P(E|D)\ P(D)}{P(E)}$$

↓ Posterior Probability

↓ Decision

(Disease likely / Disease unlikely)

4) Student Performance Prediction (Attendance, marks, Study hours)

↓ Data collection

↓ Data Preparation (Clean + Select features)

↓ Model Training (Decision Tree / Regression)

↓ Testing & Evaluation (Accuracy / Error)

↓ Prediction (High Performance / Low performance)

5)

Information Theory

↓

Uncertainty

↓

Entropy (Measures uncertainty in dataset)

↓

Feature selection

↓

Information Gain [IG = H(S) - H(S|A)]

↓

Higher Information Gain

↓

Better Feature → Better Classification

6) Machine Learning

↳ Supervised Learning

↳ Labelled Data

↳ Classification / Regression

↳ Spam Detection

↳ Unsupervised Learning
  ↳ Unlabelled data
  ↳ Clustering
  ↳ Customer segmentation
↳ Reinforcement Learning
  ↳ Reward / Penalty
  ↳ Learning through interaction
  ↳ Robotics / Game Playing

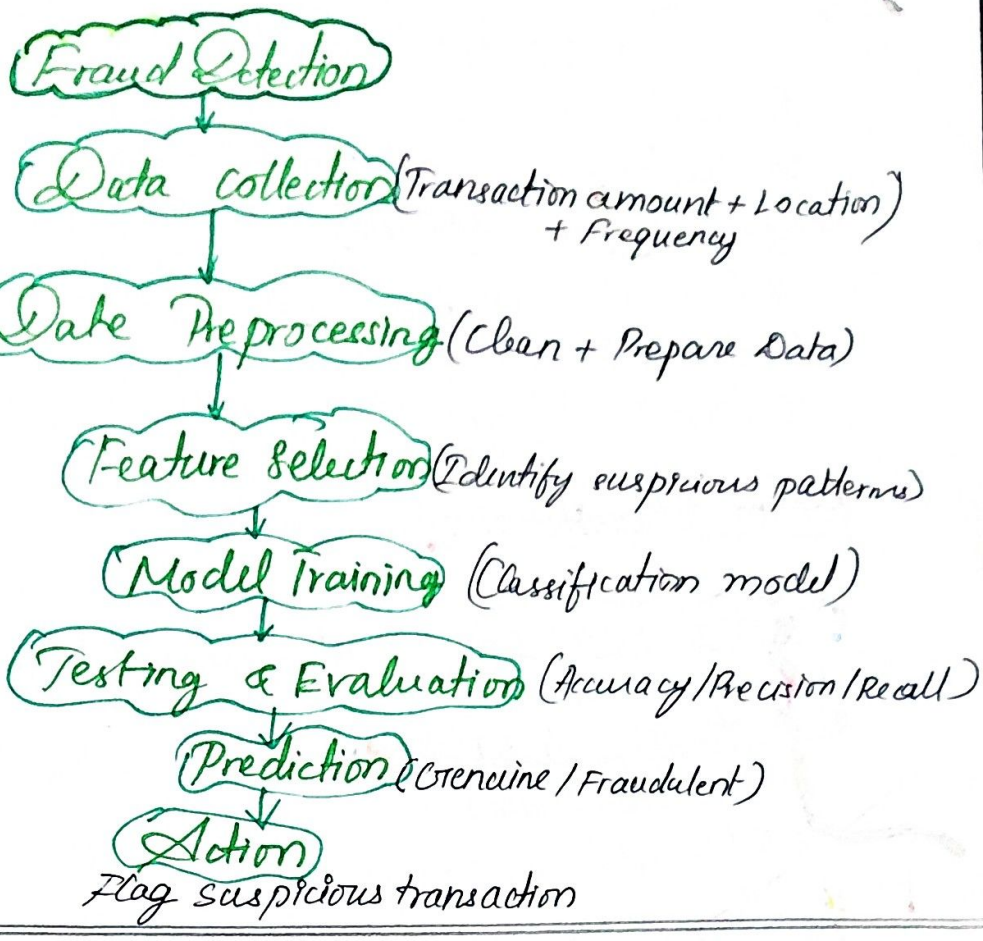
7) **Spam Classification**

- Random variables (Email features / Words)
- Conditional Probability
  - ↳ P(Word / Spam)
  - ↳ P(Word / Not Spam)
- Bayes Theorem
  - ↳ Calculate P(spam / words)
- Compare Probabilities
- Decision
  - ↳ Spam / Not spam

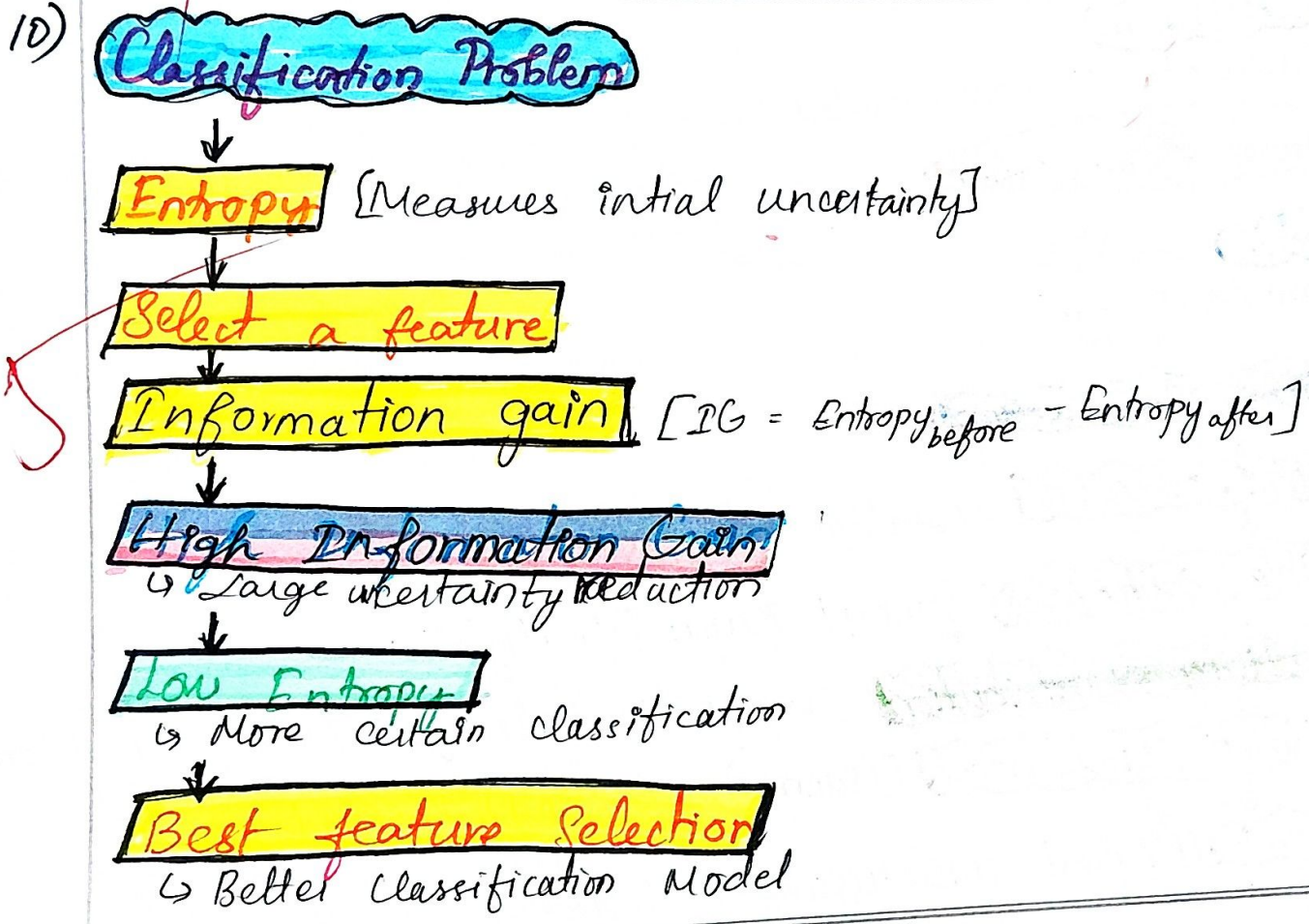
8) **Weather forecasting**

- Random variable (Weather condition)
- Possible outcomes (Sunny / Rainy / Cloudy)
- Probability Distribution (Assign probability to each condition)
- Calculate Likelihood P(Sunny), P(Rainy), P(Cloudy)
- Decision (Highest probability = Predicted weather)

9)


Fraud Detection
Data collection (Transaction amount + Location + Frequency)
Data Preprocessing (Clean + Prepare Data)
Feature selection (Identify suspicious patterns)
Model Training (Classification model)
Testing & Evaluation (Accuracy/Precision/Recall)
Prediction (Genuine / Fraudulent)
Action
Flag suspicious transaction

10)


Classification Problem
Entropy [Measures intial uncertainty]
Select a feature
Information gain [IG = Entropy before - Entropy after]
High Information Gain
↳ Large uncertainty reduction
Low Entropy
↳ More certain classification
Best feature selection
↳ Better classification Model